Looping Through Arrays

Arrays are often used with loops to iterate through all elements.

Example:

```
int i;
int numbers[5] = {10, 20, 30, 40, 50};
for (i = 0; i < 5; i++) {
    printf("%d ", numbers[i]);
}
```

Advantages of Arrays

- **Efficient Access:** Access any element in constant time using its index.
- **Memory Management:** Arrays provide contiguous memory storage, improving cache performance.
- **Simplified Code:** Reduce the need for multiple individual variables for related data.

5.2 Multidimensional Arrays

What are Multidimensional Arrays?